The 40 Hour Cabin: Master Edition (loom)

Master Edition

Donovan Zeanah

How to read this book

This printing goes along the path bedtime. Every square below is a chapter: across is a perspective, down is a level. Climb a column to follow one idea from story to activity; read a row to have the whole thing at one depth.

	The Tree	The Split	The Triangle	The Dome	The Joint	The Panel	The Raising	The Skin	The Ground	The Uses	The Count	Taking Care
Read It Together	The Tree That Fell Down	Cut It Like a Cake	The Shape That Will Not Wobble	A Ball Made of Triangles	Holding Hands Around the Corner	Forty Triangles on a Table	Up, Up, Up	A Coat for the Rain	A Good Place to Stand	What Will It Be?	Counting the Hours	Grown-Up Jobs and Helper Jobs
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Make It Yours												
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The Tree | Read It Together

The Tree That Fell Down

A storm, a pine on the ground, and a question nobody had asked



One night there was a big storm.

The wind pushed and pushed on a tall pine tree at the edge of the woods. The tree bent, and swayed, and creaked. And in the morning, when the sun came up, the tree was lying down on the ground.

Its roots were sticking up in the air, still holding on to the dirt. Its top was lost in the bushes. And in between was the trunk: long, and straight, and round, like the biggest pencil you have ever seen.

Most people who walk past a fallen tree see a problem. *It's in the way. It will rot. Somebody should cut it up for firewood.*

But the builder in this story looked at the tree and asked a different question.

What if this tree could become a house?

That sounds silly at first. Houses are made of boards from the store, aren't they? Flat, square boards, all the same.

But here is a secret about boards. Every board in the store started out as a round tree, just like this one. Somebody cut the tree down. Somebody dragged it out of the woods. A truck carried it to a mill. Big saws cut it into squares, and a lot of the round tree was cut away and thrown on a pile. Then the boards were dried,

and smoothed, and sorted, and loaded on another truck, and carried to a store, and stacked on a shelf, and sold.

That is a lot of hands, and a lot of trucks, and a lot of waiting, before a single board reaches the house.

And this tree was already here. Right where the house could be.

So the builder made a plan: no mill, no truck, no store. Just the tree, a small saw, a lot of careful thinking – and a lot of help from the shape of the tree itself.

The rest of this book is that plan.

:::box do “Try it together” Go for a walk and look for a fallen tree or a big fallen branch. When you find one:

1. Find where it is straight, and where it starts to bend or branch.
2. Wrap a piece of string around the fattest straight part, and mark the string with your finger. That is how fat the tree is *all the way round*.
3. Walk along it and count your steps from the fat end to where the straight part stops.

Take the string home. You will use it again on rung two. :::

:::box why “For the grown-up” The method in this book builds a cabin frame from trees split on the spot with one chainsaw. The reference dome is 19.42 feet across with 277 square feet of floor, framed from 120 wooden members. What matters about a tree at this stage is only three things: how fat it is at the bottom, how fat at the top of the straight part, and how long that straight part is. Everything else in the plan is worked out from those. :::

:::box wonder “Wonder question” How many different people do you think touch a board before it gets to the store? :::

The Split | Read It Together

Cut It Like a Cake

Half, half again, and half once more -- and every piece the same

First, the builder cut the long trunk into shorter pieces, each one about as long as a grown-up is tall. Each piece was a fat round log, like a giant spool of thread.

Then came the clever part.

Have you ever cut a birthday cake? You cut it right down the middle, and you have two halves. You cut each half down the middle, and you have four. You cut each of those down the middle, and you have **eight** pieces.

And here is the magic: all eight pieces are the same shape. You did not have to measure anything. You did not have to be careful about the angle. The cake was round, so cutting it in half, and half again, and half again, gives eight pieces that match – because the cake did the measuring for you.

The builder did exactly that to each round log, with a chainsaw, carefully, in long straight cuts down its length.

Half. Half again. Half again.

Eight sticks came out of every log. Each one was shaped like a long slice of cake: pointy at the middle of the log, and round on the back where the bark is. A stick shaped like that is called a **wedge**.

Most builders would never do this. They would cut the round log into square boards. But try to cut a square out of a circle, and look at what is left: four rounded corners of perfectly good wood, thrown away.

The cake cut throws almost nothing away. Every slice is used.

:::box do “Try it together” You need two things that are round – two oranges, two round slices of bread, or a round cake (the best one, because you get to eat it after).

1. Cut the first one in half, then each half in half, then each quarter in half. Count the pieces. Are they the same?
2. Now try to cut the biggest *square* you can out of the second one. Put the leftover pieces in a pile.
3. Which way wasted less? :::

:::box why “For the grown-up” The trunk is cut into sections, and each section is ripped lengthwise with a chainsaw through its centre: once to halve it, twice to quarter it, four times to make 8 sectors of 45.0 degrees each. No angle is ever measured; the round log supplies the symmetry. Split this way, a trunk keeps about 88 per cent of its wood as usable members; sawn into two-by-fours, the same log keeps about 45 per cent. The cuts are made with the saw – never with a maul or a splitting wedge – and only by an adult in full safety gear. :::

:::box wonder “Wonder question” Why do all eight pieces come out the same, even though nobody measured them? :::

The Triangle | Read It Together

The Shape That Will Not Wobble

Push a square and it leans; push a triangle and it holds

The wedges were not perfect. Some were a little bent. Some had a knot, or a bump, or one side a little fatter than the other. A tree is not a factory.

A worried builder might say: *How can I build a house out of sticks that are not perfect?*

The answer is a shape. The **triangle**.

Here is how to see it. Make a square out of four sticks, with a loose pin at each corner. Now push the top corner sideways. What happens? The square leans over. It squashes into a diamond, and then it falls flat. Nothing broke – the sticks are still the same length – but the square could not hold its shape.

Now make a triangle out of three sticks, with a loose pin at each corner. Push it. Push it harder.

It will not lean.

A triangle cannot change its shape unless one of its sticks gets longer or shorter. And a stick does not want to get longer or shorter. So the triangle holds.

That is why bridges, and swings, and roofs are full of triangles. And that is why the builder's house is made of nothing *but* triangles. When every part of a building is a triangle, the building holds its shape all by itself – even if the sticks are a little bent, or a little bumpy.

The builder liked to say it like this: *your head is not perfectly even on both sides either, and it still protects your brain just fine.*

The sticks are allowed to be a little bit wrong. The triangles make them strong.

:::box do "Try it together" Cut seven strips of cardboard or stiff paper, all about the same length, and make a hole near each end. Join four into a square and three into a triangle, with paper fasteners (or a pencil poked through) at the corners.

Push each one sideways. Which one wobbles? Now add one more strip across the square, corner to corner. What happened to the wobble? :::

:::box why "For the grown-up" A square frame is a mechanism: its corners turn and it racks over. A triangle is fixed by the lengths of its three sides, so a frame made only of triangles carries loads along its members – push and pull, end to end – rather than by bending them. That is what lets a split log with a bow, a knot or an uneven section be a structural member here. The two parts that do have to be right are the skin, because water gets in, and the ground, because everything sits on it. Later rungs return to both. :::

:::box wonder "Wonder question" How many triangles can you find holding something up, in your house or on your street? :::

The Dome | Read It Together

A Ball Made of Triangles

Twenty triangles make a ball, a few more make it round, and half of it is a house

If triangles are so strong, how do you make a whole house out of them?

Start with a ball.

Take twenty triangles, all exactly the same, and fit them together edge to edge. They fold up into a shape like a big jewel, or a twenty-sided game die. It is almost a ball – but it is a bit too pointy. You could not live in a jewel.

So the builder did something clever. Each big triangle was cut into four smaller triangles, like cutting a sandwich into four. Then every new corner was pushed out, just a little, until it touched the skin of a perfect ball.

Now the shape is much rounder. It is made of eighty small triangles, and it looks like a ball made of glass panes.

And then the builder cut the ball in half – right around its middle, like an egg cup.

Half a ball sits flat on the ground. It has a floor. It has a roof that comes all the way down to be its walls. That is a house: a **dome**.

This dome has 40 triangles. Count them in the picture if you like. And here is the best part: there are only *two* kinds.

- 10 of them have all three sides the same length.
- 30 of them have one long side and two sides that are a little shorter.

So in the whole house there are only two lengths of stick: a long one and a slightly shorter one. If you can cut two lengths, you can build the whole thing.

:::box do “Try it together” Find something in your house made of flat shapes put together into a round one: a soccer ball, a lampshade, a wooden toy. Count the shapes. Are any of them the same?

If you have a soccer ball, look for the five-sided patches. There are always twelve of them, on every soccer ball in the world. Rung six tells you why. :::

:::box why “For the grown-up” The shape is a two-frequency (“2V”) geodesic hemisphere. It begins as an icosahedron, the twenty-faced solid; every edge is halved, the midpoints are pushed out onto the sphere, and the sphere is cut at its equator. The result has 40 panels, 26 corners and only two edge lengths, in the ratio 1.1308 to one. The reference dome is 19.42 feet across and 9.71 feet tall at the centre. :::

:::box wonder “Wonder question” Why did the builder cut the ball in half, instead of living inside a whole one? :::

The Joint | Read It Together

Holding Hands Around the Corner

Three sticks, no knot in the middle, and a corner left empty on purpose

Now the builder had a pile of wedges in two lengths, and a plan for a ball of triangles. But how do you hold three sticks together at a corner?

Many dome builders use a metal star at every corner. All the sticks run into the star, and bolts hold them there. It works – but only if every stick is exactly the same, exactly straight and exactly the right length. Remember the wedges? They are *not* exactly anything. A metal star would not fit them.

So the builder did something surprising. At the corners of these triangles, **the sticks do not meet at all.**

Here is how. Hold three pencils in a triangle. Instead of making their ends touch in the corners, slide each pencil along a little, so that the end of one pencil rests on the *side* of the next one. The end of the first rests on the side of the second. The end of the second rests on the side of the third. The end of the third rests on the side of the first.

They go round and round, like children holding hands in a circle where each one holds the next one's elbow. Or like the blades of a pinwheel. And right in the corner, where you would expect the sticks to meet, there is… nothing. Just air.

Because each stick only has to rest on a flat side, it does not need to be perfect. A side is a big place to land.

And every triangle has its *own* three sticks. Where two triangles sit next to each other, two sticks lie side by side, with a thin strip of wood called a **key** pressed between them, like the filling in a sandwich.

:::box do "Try it together" Take three craft sticks, pencils or straws. Lay them on the table in a triangle so that the end of each one rests on the side of the next, going round in the same direction. Now try going round the other way. Do you get the same triangle, or its mirror image? :::

:::box why "For the grown-up" This is the pinwheel joint, and the frame is hubless: each member's butt end lands on the flat sawn side of its neighbour, and no two ends ever meet. Every panel owns all three of its members, which is why a dome of 40 panels has 120 members rather than one per edge, and why two members and a key sit in each of the 55 seams. The butt end carries the frame's one compound cut, made against a fixed setting; the other end is left long and sawn off flush once the panel is together. :::

:::box wonder "Wonder question" Why would leaving the corner empty make the building easier to make? :::

The Panel Read It Together

Forty Triangles on a Table

Build every piece flat, on one table, the same way every time

Imagine trying to build a ball of triangles up in the air, on a wobbly ladder, holding three long sticks at once and a drill in your teeth.

The builder did not want to do that. So the triangles were built somewhere much nicer: flat on a table, on the ground.

The table was special. Before any wood went on it, the builder screwed little blocks onto it in exactly the right places – blocks to hold the first stick, blocks to hold the second, blocks to hold the third, and a straight edge to guide the saw. A table like that is called a **jig**.

With a jig, nobody needs to measure anything twice. You put the first stick against its blocks. You put the second stick against its blocks. You put the third stick against its blocks. You fasten them together. You saw the long ends off along the guide. And off the table comes a triangle – exactly like the last one, and exactly like the next one.

Then you do it again. And again. Forty times.

Making the same thing over and over sounds boring. But it is the secret of how one person can build a house. The first triangle takes a while, because you are learning. By the fourth one, your hands know what to do. By the fortieth, you could almost do it with your eyes closed (please don't).

And because each triangle is finished on the ground, it can be checked, and fixed, and even painted, before it ever goes up.

:::box do "Try it together" Draw one triangle on a piece of cardboard. Push a pin into the cardboard at each corner, just outside the lines. Now lay a sheet of paper on the cardboard, and trace a triangle using the pins as your guide – then another sheet, and another.

Lay all your paper triangles on top of each other. Are they the same? Now try drawing three triangles *without* the pins and compare. :::

:::box why "For the grown-up" The jig is one flat board that holds every member against fixed stops, so the position of each stick – not a tape measure – decides the panel's shape. The dome needs 10 panels of one shape and 30 of the other, so the jig is set up twice. Repeatability beats accuracy when you need forty of something: a jig that is slightly off builds forty panels that still agree with each other, and agreeing is what makes the dome close. :::

:::box wonder "Wonder question" What is something you have done so many times that your hands know how to do it without you thinking? :::

The Raising | Read It Together

Up, Up, Up

The bottom ring first, then the next, then the very top

Forty triangles were stacked up and ready. Now it was time to make them into a house.

You cannot start at the top. The top has nothing to stand on. So the builder started at the bottom.

First the bottom ring: the triangles that touch the ground, all the way round in a circle. Then the next ring of triangles, leaning in, resting on the first. Then the next. Every ring gets a little smaller, leaning in a little more, like the rings of a beehive.

While the rings are leaning, they want to fall inward. So the builder held them up with long sticks pushed against them from inside, called **props**. A triangle goes up – a prop goes under it. Another goes up – another prop.

And then, at the very top, the last few triangles close the hole in the roof.

Here is the wonderful part. The moment the top is closed, every triangle is holding up every other triangle. Push on one, and it pushes on its neighbours, and they push on theirs, all the way round and all the way down. The dome holds itself. One by one, the props come out, and nothing falls.

The builder of this book did something even more daring. Instead of building on the ground, the builder strung strong ropes between big living trees, like a giant spider web in the air, and lifted each triangle up into the web with a winch – a machine that winds a rope around a drum. The triangles were joined together up in the air, and the whole dome hung between the trees, like a lantern.

:::box do “Try it together” Stand up some books in a ring on the floor, leaning a little inward. Do they fall? Now lean a second ring on top of the first. What did you need to keep them from falling? Those are your props. :::

:::box why “For the grown-up” A dome goes up course by course from the base ring, propped from inside until the crown closes; once it closes the shell carries itself and the props come out. The author’s own method builds in the air instead: a web of wire rope between living trees, a 1,500-pound 12-volt winch, and each triangle lifted flat in a harness and joined in place. Raising anything heavy is grown-up work, and nobody – ever – stands under a load that is hanging. :::

:::box wonder “Wonder question” Why does a dome hold itself up once the very top is closed, but not before? :::

The Skin | Read It Together

A Coat for the Rain

One coat that keeps the water out, and soft warm layers underneath

The dome was standing. But if you stood inside it, you could look up through every triangle and see the sky. The wind blew straight through. And when it rained – splash.

A house needs a coat.

Water is sneaky. It runs downhill, it creeps into cracks, and if it gets inside wood and stays there, it makes the wood soft and rotten. A house can be strong, and straight, and beautiful, and still be ruined by a little water that got in and never got out. So the builder thought about water more carefully than anything else.

The answer was one coat that keeps out every drop of rain. Just one. Like a raincoat, or a shower cap pulled down over the whole dome. Everything else goes *under* the coat, where it stays dry.

And under the raincoat? Warm, soft layers – like the quilts on a bed. The builder's quilts can be sewn from old clothes, the ones too worn out to wear. Each layer makes the house a bit warmer. When winter comes, add another quilt. The next winter, add another. A house that gets warmer every winter, like a person putting on one more hat.

There is one more sneaky kind of water: the kind that comes from inside. When it is warm in the house and cold outside, your breath makes tiny drops on the cold walls, just like on a cold window. So the builder left the house a way to breathe, and a place for that water to drip out.

:::box do "Try it together" Take two pieces of paper and a tray. On the first, lay three strips of paper side by side with their edges just touching, and tape them. On the second, lay three strips *overlapping* like roof shingles, the top one over the middle one over the bottom one. Tilt each like a roof and sprinkle water from the top. Which one lets water through to the tray? :::

:::box why "For the grown-up" The skin, not the frame, is the part of a dome that has to be right, because water is what ruins wooden buildings. This design uses exactly one watertight layer – a cap over the whole shell – with insulation underneath it. The insulation is quilted from recycled clothing: about 132 t-shirts a layer, each layer adding R-1.6, added one winter at a time. Moisture from inside is handled on purpose, by letting the shell breathe and draining water at the rim. :::

:::box wonder "Wonder question" Where does the water on the inside of a cold window come from? :::

The Ground · Read It Together

A Good Place to Stand

A house is only as steady as the ground under its ring

Have you ever sat at a table with one short leg? Every time you lean on it, it wobbles. Clunk. Clunk.

A dome is a little like that table. All of its weight comes down through its bottom ring: a circle of 10 corners sitting on the ground. If the ground under one corner is lower than the others, or soft, or sinks when it rains, the whole dome leans a little – and a dome that leans a little at the bottom leans more at the top.

So the builder said: *the frame is allowed to be a little bit wrong. The ground is not.*

The ground under the ring has to be level, all the way round. And it has to be strong enough not to sink. There are lots of ways to make good ground: a ring of flat stones, a bed of packed gravel, a wooden deck on short posts, or a concrete floor. The builder checks the ring by measuring straight across it, corner to corner, in several directions. If every measurement is the same, the ring is round. If one is different, the ring is telling you where it is wrong, while it is still easy to fix.

And here is an idea that is bigger than one house. What if one family owned a good flat piece of ground with water and electricity ready, and another family brought their dome and set it down there? The ground stays; the dome can move on someday. The builder calls that ground a **pad**.

:::box do “Try it together” Make a ring of ten paper cups or toy blocks on a pillow, and set a plate on top. Now do the same on a flat table. Push gently on each plate. Which one rocks?

Then measure across your table ring with a string in three different directions. Are the three the same? :::

:::box why “For the grown-up” The base ring is the one place in the method where error does not get absorbed – it accumulates, because every course above stands on whatever circle the ring actually is. That is why the ring is checked across opposite corners when it is set and again as the first courses go on. Footings range from gravel to a slab; the options and their costs are on rung three. A pad is serviced ground built once and shared or rented, so the dome on it can be moved. :::

:::box wonder “Wonder question” What is the steadiest thing you have ever stood on, and why was it steady? :::

The Uses | Read It Together

What Will It Be?

The same round frame can be a bedroom, a workshop, a greenhouse or a sauna

Here is a riddle. What is round, made of triangles, and can be a bedroom, a workshop, a greenhouse, a sauna, a garage, a studio, a gym and a cold room for keeping food?

The answer: the same dome.

Every triangle in the dome is an opening. What you put *in* the opening decides what the building is. Put windows in, and it is full of light. Put a door in, and you can walk through. Put clear panels in the top, and plants can grow inside. Put a little chimney through one, and you can have a wood stove. Put in solid, warm panels, and it is a cosy bedroom.

So the builder thinks of the bare frame like a seed, or like a baby cell in your body that has not decided yet whether it will be skin, or bone, or brain. Scientists call a cell like that a **stem cell**. The frame is the stem cell. The panels decide what it grows into.

And if you change your mind? Take a panel out and put a different one in. The greenhouse can become a bedroom. The bedroom can become a workshop. The frame does not mind. It stays exactly the same.

There is one more piece the builder treats as special: a column right in the middle of the floor that carries water, power and drains. It is the most expensive part, so it is built to unbolt and move with the family to the next dome, the way you would take your favourite chair when you move house.

:::box do “Try it together” Draw a big circle on a piece of paper. That is your dome’s floor. Decide what your dome is for, and draw what goes inside it. Then turn the paper over, draw another circle, and make the same dome into something completely different. :::

:::box why “For the grown-up” The “stem cell” is the project’s name for the bare frame, skin, core and one blank panel, with the building’s purpose deferred to the panels hung in its forty openings. The catalogue prices 15 fit-outs on the same body, from 12,036 dollars for the bare article to 25,511 dollars for the most fitted. Many of them are utility buildings – storage, a workshop, a greenhouse, a studio – which is usually where a family’s first dome earns its keep. :::

:::box wonder “Wonder question” If you could have one round room all to yourself, what would it be for? :::

The Count | Read It Together

Counting the Hours

How long a house takes, and why counting honestly matters

How long does it take to build a house?

For most houses, the answer is: a long time, a lot of people and a lot of money. The builder of this book wanted to know exactly how long *this* house would take, so they counted.

They counted the hours of cutting wedges at the log: about 16 hours. They counted cutting the ends of the sticks, and building the triangles on the jig, and lifting them up into a dome. And then – because the first time you build anything, things go wrong – they added extra hours just for mistakes.

All together the frame came to about 37.5 hours. A grown-up's working week is 40 hours. So the frame of the house fits inside one week of work. That is why this book is called *The 40 Hour Cabin*.

But the builder was careful to say what the forty hours do *not* include: cutting down the trees, putting the coat on, the floor, the door, the windows. Those come after, and they take longer. All of it together is about 67 hours of building.

The builder was also careful about something else. Some of those numbers were *measured* – timed with a clock while the work was really done. Others were *guesses*, good guesses, but guesses. In this book, every number says which kind it is. Because a number that pretends to be measured when it was guessed is a kind of fib, and you cannot build a house on fibs.

:::box do "Try it together" Get a clock or a timer. Time how long it takes to cut out and tape together 5 paper triangles. Now work out how long forty would take: forty is 8 lots of 5. Was your guess before you started close to the real answer? :::

:::box why "For the grown-up" The forty-hour clock covers the frame, from wedges to standing: 16.0 hours of splitting at the measured rate of 30 wedges per 4-hour session, then cutting, panels and raising as estimates, plus the owner's own error allowance of 7.5 hours. Felling, the skin, the floor and the openings are not on that clock; the whole build is about 67 hours. Measured and estimated figures are labelled as such everywhere in this book. :::

:::box wonder "Wonder question" Why does it matter whether a number was measured or guessed? :::

Taking Care | Read It Together

Grown-Up Jobs and Helper Jobs

The saw is a grown-up's job; the counting, sorting and painting are yours

Building a house is a team job. And on every good team, everybody has the right job.

Some jobs are grown-up jobs. The chainsaw is a grown-up job: it is sharp, and fast, and very loud, and it does not care whose leg is in the way. Cutting down a tree is a grown-up job too, and even grown-ups who have done it a hundred times are careful every single time. Climbing ladders, lifting heavy triangles, working with electricity and water pipes – all grown-up jobs.

When a grown-up uses the chainsaw, they wear special clothes. A hard hat with a screen over the face. Ear protectors, because the saw is loud enough to hurt your ears. Thick gloves. Strong boots. And special trousers called **chaps**, full of tough threads that can stop a chain if it ever touches them.

And there is one rule the builder never, ever breaks: **nobody uses the saw alone.** There is always somebody else there – standing well back, watching, ready to help.

So what can a helper do? Lots!

A helper can count the sticks and sort them into piles by length. A helper can mark each stick with a crayon so everybody knows which pile it belongs to. A helper can hold the end of the measuring tape. A helper can draw the plans, time the work and write down the numbers. A helper can build the paper models that show everybody what the real house will look like. A helper can help oil the wood – with gloves on. And a helper can ask the best questions of all.

The most important job on any team is the one that keeps everybody safe.

:::box do “Try it together” Make two lists on one piece of paper. On one side, write “Grown-up jobs”. On the other, “Helper jobs”. Go through the twelve chapters of this rung and put every job you can think of on one list or the other. Is there anything you are not sure about? Ask! :::

:::box why “For the grown-up” Children stay well outside any felling zone – at least two tree-lengths from a tree being cut – and away from a running saw. Chainsaw work needs a helmet with a visor, hearing protection, gloves, boots and chainsaw chaps, and a second person present every time. Oily rags from finishing can catch fire by themselves: they go straight into a metal tin of water. Before anyone sleeps in a building made this way, an engineer and the local building department must have seen it. Rung three turns all of this into a written plan. :::

:::box wonder “Wonder question” What is the bravest thing a person can say on a building site? (Hint: it is “wait”.) :::